

Anime line art colorization by region matching using region shape

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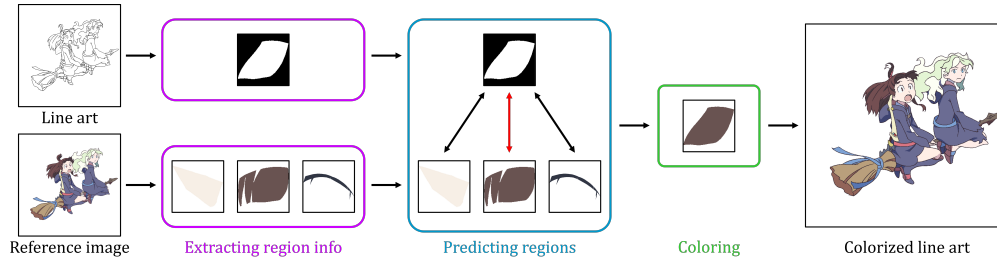


Figure 1: Overview of our method.

Abstract

The coloring process in anime (Japanese animation) production constitutes a critical yet time-consuming aspect of creating animated works. This study introduces a colorization system for anime line art using a reference image. The proposed system colors line art based on the features of corresponding regions in the reference image, significantly reducing the time required for the coloring task.

CCS Concepts

• Computing methodologies → Computer graphics.

Keywords

Anime, Colorization, Neural Network

ACM Reference Format:

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1 Introduction

The coloring phase in anime production is essential but demands substantial time and effort. Interviews with Studio Trigger [Trigger, Inc. 2024], a Japanese animation studio, revealed three key insights into this process. First, coloring a single line art can take up to 30 min, resulting in considerable time spent by the coloring

staff. Second, the anime industry lacks automated coloring systems, necessitating manual color of nearly identical line art. Finally, implementing automated coloring systems could significantly benefit the coloring staff. Additional findings from the interviews indicated that anime line art differs from live-action line images in that it features two types of lines—those present after coloring and those absent. Furthermore, anime images and their line art are segmented into numerous regions, often exceeding 100, with each region typically assigned a single color, except in specific cases.

The graph correspondence-based coloring [Do et al. 2021] can automatically color manga (Japanese comics) using reference images. This method relies on the principle that manga consists of numerous distinct regions. It employs two indices—shape and inter-region nodes—to align the line art with regions in the reference image. However, this approach cannot address superfluous lines absent from the final colored image, rendering it unsuitable for anime applications. Also, area correspondence is calculated based on the node between regions, there is a concern that the calculation time will be long. Conversely, the neural network-based coloring method [Cao et al. 2023] can color-handle line art and reference images with dissimilar compositions. Nevertheless, this method is unsuitable for anime because a single region may contain multiple colors.

This study introduces a novel colorization system for anime line art using a reference image. The primary objective is to minimize the time and effort involved in the coloring process.

2 Our Approach

Fig. 1 presents an overview of our proposed method. The approach consisted of three steps. The first step involved extracting region information from line art and reference images. The second step focuses on predicting regions based on the extracted information. The final step included coloring the line art and removing extraneous lines to produce the completed anime image. Our method

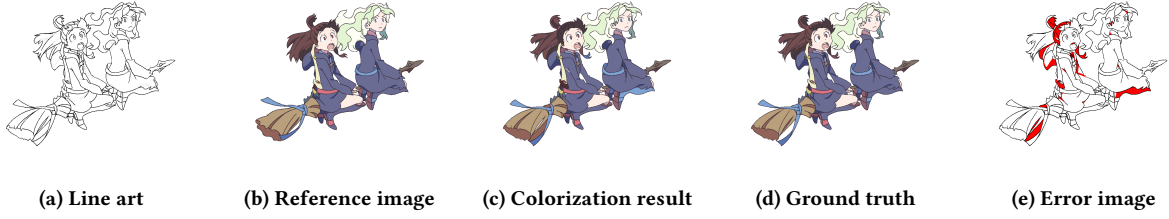


Figure 2: Example of colorization result. Line art is the next frame of reference image.

did not directly generate colored images but produced images of many regions, each assigned a single color. The region estimation method in the second step was inspired by the technique developed by Do et al. [Do et al. 2021]

In the initial step, our method gathered information about each region in both images, including the centers of gravity, areas, and shape features. The centers of gravity and areas were represented as numerical values ranging from 0 to 1, while shape features were encoded in 512-dimensional vectors derived from ResNet34. [He et al. 2016]

A method in the second step used three indices — the center of gravity, area, and shape feature — to identify the corresponding region in both images based on a cost function. The cost function assessed the similarity between the i -th region in the line art and the j -th region in the reference image. Our method identified the corresponding j -th region that minimized the cost function for the i -th region. Cost function F_1 determines the corresponding region in large regions, and cost function F_2 determines the corresponding region in small regions.

$$F_1(i, j) = \exp(\alpha_1 \cdot D_{shape}) \cdot \exp(\beta_1 \cdot D_{area}) \cdot \exp(\gamma_1 \cdot D_{center})$$

$$F_2(i, j) = \exp(\alpha_2 \cdot D_{shape}) \cdot \exp(\beta_2 \cdot D_{area}) \cdot \exp(\gamma_2 \cdot D_{point})$$

where D_{shape} represents the sum of the absolute differences between the shape feature vectors of the i -th and j -th regions. Lower values of D_{shape} indicate more similar shapes. Furthermore, D_{area} represents the absolute difference in the areas of the two regions, with lower values indicating more similar areas. D_{center} denotes getting low when both regions are similar areas. D_{point} denotes the Manhattan distance between the centers of gravity of the two regions. D_{point} denotes the total difference between the distance from the i -th region to the region associated with F_1 and the distance from the j -th region to the region associated with F_1 . Parameters α_1 and α_2 were set to 0.001, while β_1 , β_2 , γ_1 and γ_2 were set to 1.

The final stage involved coloring the line art and removing superfluous lines to complete the anime image. This process involved filling each region in the line art with the color corresponding to the presumed region in the reference image. After colorization, our method eliminated unnecessary lines that do not appear in the anime image. If a pixel was located at the edge of the image, its value was replaced using the values of three or five adjacent pixels.

3 Result

We evaluated our method using a dataset [National Institute of Informatics 2022] comprising real animated works [TRIGGER et al. 2015], and generated 62 images from 31 cuts. These line arts are

the previous one frame and the next one frame of the reference images in the same cut. We compared the accuracy rate of our proposed method with that of existing method (AnT) [Casey et al. 2021], and used software [Cadmium 2021] as existing method. The proposed method was able to colorize all 62 images, but AnT was unable to colorize five images. The average region-based accuracy rate for the 57 images was 77.18% for the proposed method and 75.25% for AnT. Fig. 2 shows an example of colorization result of our proposed method. The results demonstrated that the large regions were accurately colored, whereas smaller regions were occasionally assigned incorrect colors. The accuracy rates were 99.72% on a pixel basis and 84.21% (128 out of 152 regions) on a region basis.

4 Conclusions

In this paper, we propose a colorization method for anime line arts. Users do not need to prepare large amounts of data obtained permission from copyright owners, because our method does not use machine learning technique. When the line art and the reference image have similar composition our method can color many regions correctly. We confirmed that our method tends to color large areas correctly.

Acknowledgments

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